

A Fixed Point:

If g is defined on $[a, b]$ and $g(p) = p$ for some $p \in [a, b]$, then the function g is said to have the fixed point p in $[a, b]$.

- The fixed-point problem turns out to be quite simple both theoretically and geometrically.
- The function $g(x)$ will have a fixed point in the interval $[a, b]$ whenever the graph of $g(x)$ intersects the line $y = x$.

Given such a function, $f(x)$, we now construct an auxiliary function $g(x)$ such that

$$p = g(p)$$

whenever $f(p) = 0$ (this construction is not unique).

The problem of finding p such that $p = g(p)$ is known as the **fixed point problem**.

Fixed-Point Iteration

Start with an initial guess x_0 of the root, and form a sequence $\{x_k\}$ defined by

$$x_{k+1} = g(x_k); k = 0; 1; 2;$$

If the sequence $\{x_k\}$ converges, then $\lim_{k \rightarrow \infty} x_k = p$ will be a root of $f(x) = 0$.

The question therefore rises:

Given $f(x) = 0$, how to write $f(x) = 0$ in the form $x = g(x)$, so that starting with any x_0 in $[a, b]$, the sequence $\{x_k\}$ defined by $x_{k+1} = g(x_k)$ is guaranteed to converge?

Fixed-Point Iteration

- The simplest way to write $f(x) = 0$ in the form $x = g(x)$ is to add x on both sides, that is **$x = f(x) + x = g(x)$**
- But it does not very often work. To convince yourself, consider the following example

$$f(x) = x^3 - 6x^2 + 11x - 6 = 0.$$

$$\text{Define } g(x) = x + f(x) = x^3 - 6x^2 + 12x - 6$$

- We know that there is a root of $f(x)$ in $[2.5, 4]$; namely $x = 3$

Fixed-Point Iteration

- Let's start the iteration $x_{k+1} = g(x_k)$ with $x_0 = 3.5$
- Then we have

$$x_1 = g(x_0) = g(3.5) = 5.3750$$

$$x_2 = g(x_1) = g(5.3750) = 40.4434$$

$$x_3 = g(x_2) = g(40.4434) = 5.6817 \times 10^4$$

$$x_4 = g(x_3) = g(5.6817 \times 10^4) = 1.8340 \times 10^{14}$$

- **The sequence $\{x_k\}$ is clearly diverging**

Convergence

- The following theorem gives a sufficient condition on $g(x)$ which ensures the convergence of the sequence $\{x_k\}$
- **Fixed-Point Iteration Theorem:** Let $f(x) = 0$ be written in the form $x = g(x)$. Assume that $g(x)$ has the following properties:
 1. For all x in $[a; b]$; $g(x)$ in $[a; b]$; that is $g(x)$ takes every value between a and b .
 2. $g'(x)$ exists on $(a; b)$ with the property that there exists a positive constant $r < 1$ such that $|g'(x)| \leq r$; for all x in $(a; b)$.

Convergence

- Then

(i) there is a unique fixed point $x = p$ of $g(x)$ in $[a; b]$.

(ii) For any x_0 in $[a; b]$, the sequence $\{x_k\}$ defined by

$$x_{k+1} = g(x_k); k = 0, 1, \dots$$

converges to the fixed point $x = p$; that is to the root of $f(x) = 0$

Uniqueness Result

Let $g \in C[a, b]$ and $g(x) \in [a, b]$ for all $x \in [a, b]$. Further if $g'(x)$ exists on (a, b) and

$$|g'(x)| \leq k < 1, \quad \forall x \in [a, b],$$

then the function g has a unique fixed point p in $[a, b]$.

Example

Let $f(x) = x^3 - 6x^2 + 11x - 6 = 0$. Choose $a = 2.5$, $b = 4$.

- Let's write $f(x) = 0$ in the form $x = g(x)$ as follows:

$$x = \frac{1}{11}(-x^3 + 6x^2 + 6) = g(x)$$

- Then

$$g'(x) = \frac{1}{11}(-3x^2 + 12x)$$

- It is easy to verify graphically and analytically that $|g'(x)|$ is less than 1 for all x in $(2.5; 4)$, (note that $g'(4) = 0$; and $g'(2.5) = 1.0227$).

Example

- So, the iteration $x_{k+1} = g(x_k)$ should converge to the root $x = 3$ for any x_0 in $[2.5, 4]$. This is verified as follows:

x_0	$=$	3.5
$x_1 = g(x_0)$	$=$	3.3295
$x_2 = g(x_1)$	$=$	3.2367
$x_3 = g(x_2)$	$=$	3.1772
$x_4 = g(x_3)$	$=$	3.1359
$x_5 = g(x_4)$	$=$	3.1059
$x_6 = g(x_5)$	$=$	3.0835
$x_7 = g(x_6)$	$=$	3.0664
$x_8 = g(x_7)$	$=$	3.0531
$x_9 = g(x_8)$	$=$	3.0427
$x_{10} = g(x_9)$	$=$	3.0344

The sequence is clearly converging to the root $x = 3$

Example

Find a root of $f(x) = x - \cos x = 0$ in $[0, \frac{\pi}{2}]$

- Let's write $x - \cos x = 0$ as $x = \cos x = g(x)$. Then $|g'(x)| = |\sin x| < 1$ in $(0, \pi/2)$. Hence the iteration method is applicable.

$$x_0 = 0$$

$$x_1 = \cos x_0 = 1$$

$$x_2 = \cos x_1 = 0.54$$

$$x_3 = \cos x_2 = 0.86$$

$$\vdots$$

$$x_{17} = 0.73956$$

$$x_{18} = \cos x_{17} = 0.73955$$

Example 1. Find a root of $f(x) = x^3 + x^2 - 1 = 0$ correct to 3-decimal places.

- Rewrite the given equation as $x = \phi(x)$
- Given $x^3 + x^2 - 1 = 0 \Rightarrow x^2(x+1) = 1$

$$\Rightarrow x = \frac{1}{\sqrt{x+1}} = \phi(x)$$

Now
$$|\phi'(x)| = \left| \frac{-1}{2(x+1)^{3/2}} \right| < 1$$

Hence the iteration method is applicable.

Solution cont.

- Let $x_0 = 0.75$. Therefore

$$x_1 = \phi(x_0) = \frac{1}{\sqrt{0.75 + 1}} = 0.7559$$

$$x_2 = \phi(x_1) = 0.75465$$

$$x_3 = \phi(x_2) = 0.754924$$

$$x_4 = \phi(x_3) = 0.754869$$

Therefore 0.7549 is an approximate root of the given equation.

Example

- Find the root of $x^2 + \ln(x) - 2 = 0$, which lies between 1 and 2, by fixed-point iteration method, correct to four decimal places.

Solution: Let $f(x) = x^2 + \ln x - 2 = 0$. Now, $f(1) = -1 < 0$, $f(2) = 2.69 > 0$. Therefore, one root lies between 1 and 2.

We write the equation as:

$$x = \sqrt{2 - \ln x} = \phi(x), \quad \phi'(x) = -\frac{1}{2x\sqrt{2 - \ln x}}$$

$$\therefore |\phi'(x)| < 1 \text{ as } \text{Max } [|\phi'(1)|, |\phi'(2)|] = \text{Max } [0.35, 0.21] = 0.35 < 1.$$

$\therefore x = \sqrt{2 - \ln x} = \phi(x)$ gives us a convergent sequence of iteration.

We take $x_0 = 1$.

	n	x_n	$\phi(x_n)$
	0	1	1.4
	1	1.4	1.29
	2	1.29	1.32
	3	1.32	1.312
	4	1.312	1.3147
	5	1.3147	1.3139
	6	1.3139	1.31415
	7	1.31415	1.31408
	8	1.31408	1.31410
Check	9	1.31410	1.31409

Thus, **1.3141** is the root of the equation between 1 and 2, correct up to four decimal places.

Example 7. Find the root of the equation $3x - \cos x - 1 = 0$, by the iteration method, correct to four significant figures.

Solution. Let $f(x) = 3x - \cos x - 1$. As $f(0) = -2 < 0$, $f(1) = 1.45 > 0$, there is one root of $f(x) = 0$ between 0 and 1.

We re-write the equation as

$$x = \frac{\cos x + 1}{3} = \phi(x), \therefore \phi'(x) = -\frac{\sin x}{3}.$$

Therefore, $|\phi'(x)| < 1$, as $|\sin x| < 1$ and the convergence criteria is satisfied.

As a initial value, we take $x_0 = 0$. The successive approximations of the root are computed in tabular form as follows:

n	x_n	$\phi(x_n)$
0	0	0.6
1	0.6	0.61
2	0.61	0.606
3	0.606	0.6073
4	0.6073	0.60706
5	0.60706	0.60711
6	0.60711	0.60710

Thus, 0.6071 is a root of the equation, correct up to four significant figures.